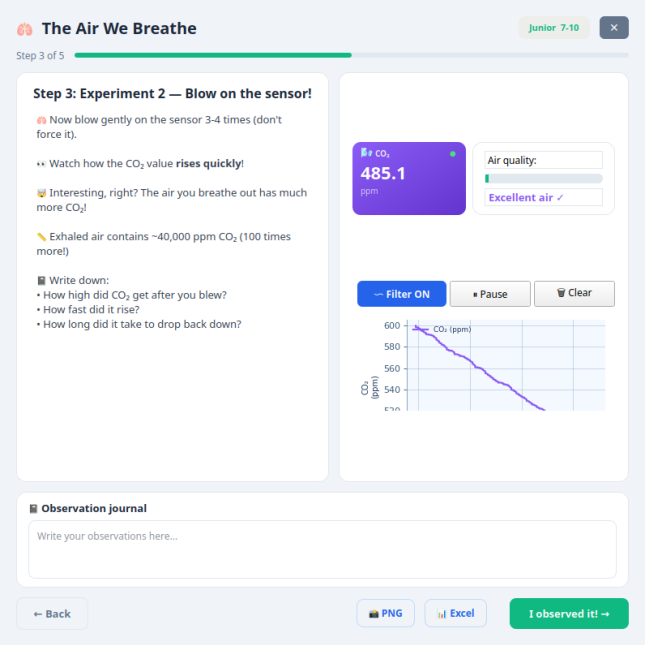
The Air We Breathe

Making an invisible gas visible — and discovering that the recovery is the interesting part

Level: Junior — ages 7–10 (Romanian school years II–IV) • **Time in class:** 15 min, or 3 short lessons in a week • **Sensors:** SCD41 (carbon dioxide, temperature, humidity)



The activity screen. The values in the picture are simulated; your own will differ.

Before you start

Who it is for

Junior — ages 7–10 (Romanian school years II–IV). The numbers decide how far you go: the classroom values, 400–2000 ppm, are four-digit numbers and therefore belong to **year III** (working within 0–10 000). In year II children work below 1000, so they compare the value at the start with the one after airing and use the words on the screen rather than the large figures. **The 40 000 ppm of exhaled air** belongs to year IV, but it is worth saying aloud at any age — not as a number to write down, but as an astonishing order of magnitude.

Assumed knowledge

Comparing two numbers. **Decimals are not required** — no number in this activity has a decimal point. Reading a graph that rises and falls helps, but it is learned here.

Time in class

Short: 15 min. Full: three short lessons in one week — 20 min, 40 min and 20 min — about an hour of actual time.

Teacher preparation

10 min — **check that the window opens** and that you are allowed to open it; power up a box and leave it two minutes to see the room's value; write the thresholds on the board; and decide in advance your hygiene rule for blowing at the sensor.

Sensor settling time	**30 seconds** after power-up before the readings can be trusted — the first ones are ignored entirely. After that the sensor produces a new value **about every five seconds**, not continuously, so the graph climbs in steps. **The recovery after a breath takes one to three minutes**, and that is the most interesting part: do not rush on.
Sensors and boxes	SCD41. It gives carbon dioxide, but also temperature and humidity, and both rise when somebody breathes on it — a useful clue. One box is enough for the whole class if you work in a semicircle with the screen projected; ideally, one box per 4–6 children. **CO_2 needs recent firmware**: 2.2.0 or later. A box still on the old USB-only firmware (`serial_only`) is recognised, and the app says to flash the WiFi firmware, which works over USB too.

Three routes

Short	15 min: written prediction, the reference values on screen, the room's air, the breath and the recovery, the window opened and the conclusion. These are the steps in the app, which also asks for the written prediction before measuring.
Full	Three short lessons: lesson 1 the short route, unhurried (20 min); lesson 2 **logging a whole lesson** (40 min); lesson 3 the air map — corridor, hall, outdoors, classroom — and ranking them (20 min).
Project	Extension: the box and a small plant in a large jar, to see what the plant does to the air — the bridge to the senior photosynthesis activity.

1. Where this fits in the curriculum

Children know that they breathe. What they have not met is that *exhaled air is different* and that the difference can be measured, even though nothing can be seen. Here the sensor does something no teaching material can: it shows an invisible gas, on a graph, in real time. The activity fits:

- **Mathematics and exploring the environment** (years I–II) — objects and phenomena in the immediate environment, air, and measuring with an instrument. At this level two *states* are compared — before and after airing — using the words on the screen.
- **Natural sciences** (years III–IV) — **human respiration**, air as a mixture, what plants need, and the link between them. This is the most direct entry: the syllabus asks for respiration as a process, and here it has a measurable result.
- **Mathematics** (a subject in its own right from year III) — reading and comparing four-digit numbers, within 0–10 000, with *real* numbers: 600 ppm in an aired room and 1800 after an hour with the windows shut. Subtraction gives the rise.
- **Citizenship and personal development** — **who opens the window, and when**. The activity ends naturally with a class rule, decided by the children on the basis of their own data, and this is one of the few occasions on which a measurement leads directly to a decision they can take themselves.
- **Cross-curricular** — the patience to watch a line come down slowly, and a first encounter with the idea of a *threshold*: not "a lot" or "a little", but above or below an agreed value.

The core skill is **reading a change over time**. Children of this age do not need rates; they need to see that a line rises when something happens, falls when something else happens, and that *they* caused both.

2. Time plan

The in-app wizard fits into a quarter of an hour. The extended version is far stronger, though, because it lets children watch their own classroom's CO_2 climb over a whole lesson, and it costs

about an hour of class time spread across a week.

Route	Time	What happens
Short	**15 min**	Written predictions (3), the introduction on screen (2), the room's air (3), breathing at the sensor and watching the recovery (5), the window opened, the **How sure are we?** step read in passing, and the conclusion (2). These are the steps in the app; the second breath that step asks for is done in lesson 1.
Lesson 1	20 min	The predictions and the two wizard experiments, plus the window. Each group records the room value, the peak after the breath, and how long the recovery took, then, at the **How sure are we?** step, breathes a second time, the same way, and compares the two peaks.
Lesson 2	40 min	**Logging a whole lesson.** One box is left running in the middle of the room from the moment the class comes in. The children get on with something else. In the last ten minutes the class looks at the graph together, finds the slow climb, then opens the windows and watches the fall. **This is the most convincing thing in the entire topic.**
Lesson 3	20 min	Where is the air freshest? Groups measure the corridor, the hall, outdoors by the door and the classroom, and rank them. Discussion of who is responsible for airing the room and when.

The recovery is the interesting part — do not rush on

The sensor needs about **30 seconds** after power-up before its values can be trusted, and then produces a new value **about every five seconds**, not continuously. The graph therefore climbs in steps, and that is normal.

In the breathing experiment, **the rise appears within a few seconds but the fall back takes one to three minutes** — and that fall is scientifically the more interesting half. Do not move on.

Fill the wait with a task, announced in advance: one child reads the number aloud every fifteen seconds so that the whole class hears it dropping; the others draw the shape of the line on their sheet — the sudden rise and the slow descent. **Watching a number crawl back to where it started is, in itself, a very good thing to see at nine.** Be honest, though, about where the slowness comes from: the breath has spread through the room within seconds, but *the sensor* needs time for the gas to leave its little chamber, and it measures only once every five seconds (see the scientific background). The real airing of the room is what you see at the window.

3. Materials

Item	Per group	Notes
MagicBox with SCD41	1	The CO_2 sensor. It also reports temperature and humidity, which both rise when someone breathes on it — a useful clue.
Laptop with the app	1	USB cable or WiFi; for CO_2 the box needs firmware $\geq$ 2.2.0.
A window that opens	1 per room	Essential for the third experiment. **Check before the lesson** that it opens and that you are permitted to open it.
Empty plastic drinks bottle, 500 mL	1	Breathe into it, cap it, then uncap it beside the sensor. A tidier and more hygienic version of breathing directly, and it lets a breath be kept and used later.
Balloon	1–2	Blow it up, then release the air slowly towards the sensor. Children love this and it gives a big, controllable puff.
Large jar or clear box with a lid	1	With the sensor inside and a small potted plant, for the "what do plants do?" extension. The lid must close well: a jar that lets air in shows nothing reliable.
Small potted plant	1	Optional, for the link to photosynthesis. Ivy, mint or basil work well.
Piece of cardboard	1	Used as a fan to move fresh air over the sensor and speed up the recovery — the simplest way to show that ventilation is just moving air.
Stopwatch or the class clock	1	For timing the recovery.
Sticky labels	4	To name the measuring locations for lesson 3.
Coloured pencils	1 set	For drawing the shape of the line.

Preparation the day before. Confirm that the window opens and look at the weather: a gale makes the window experiment chaotic, and a still, mild day makes it perfect. Power up a box and leave it two minutes to confirm you get a sensible number — somewhere between 400 and 900 ppm in an aired room. Air the room thoroughly before the lesson if you want a low, clean starting value, or **keep it deliberately closed overnight** if you want to open the lesson with a dramatically high one. Write the thresholds large on the board, because the children will refer to them constantly. And decide your hygiene rule for breathing at the sensor.

4. Scientific background for the teacher

What is in the air, and what breathing changes

Dry air is about 78% nitrogen and 21% oxygen. Everything else — argon, carbon dioxide and the rest — fits into the last one per cent, and carbon dioxide is only about 0.042% of the air. That is why the unit is **parts per million**: 420 ppm means 420 molecules of CO_2 in every million molecules of air. It is a tiny fraction of a tiny fraction, and yet it is enough to matter both for the room and for the planet.

Breathing does not remove all the oxygen and replace it with carbon dioxide. Inhaled air is about 21% oxygen and 0.04% CO_2; exhaled air is still about 16% oxygen but around 4% CO_2 — that is 40 000 ppm, which is where the app's "about a hundred times more" comes from. Aerobic respiration in every cell of the body makes the difference:

$$C_6H_{12}O_6 + 6\,O_2 \rightarrow 6\,CO_2 + 6\,H_2O + \text{energy}.$$

Note that water appears on the right-hand side too — which is why exhaled air is also warm and humid, and why the temperature and humidity on screen twitch upwards at the same instant as the CO_2. **If a child asks whether the sensor might be responding to the warmth rather than the gas, that is an excellent question**, and the answer is the bottle experiment: a breath that has had time to cool still produces the CO_2 spike.

4.1 How the sensor sees an invisible gas

The SCD41 is a **photoacoustic NDIR** sensor: non-dispersive infrared, heard rather than seen. Inside is a small chamber lit by an infrared source that flashes on and off. Carbon dioxide molecules absorb infrared strongly at a wavelength near 4.26 micrometres, while nitrogen and oxygen do not. At every flash the CO_2 in the chamber takes up a little energy, warms and expands, so the flashes become a faint sound, which a tiny microphone picks up. The more CO_2 there is in the chamber, the more infrared is absorbed and the louder the sound; the absorbed part follows

$$I_{\text{abs}} = I_0\left(1 - \mathrm{e}^{-\alpha c L}\right),$$

where c is the concentration and L the path length. The electronics invert that and report ppm.

Two things follow, and both matter in class. First, the gas has to get into the chamber physically, by diffusion through a membrane, and the sensor completes a measurement **only once every five seconds**. Together these give a response time of the order of a minute (the SCD41 data sheet gives about 60 s to reach 63% of a step). **That is where the slow recovery after a breath comes from**, not from the room being aired: the plume of exhaled air disperses within seconds, but the sensor "forgets" gradually. The rise looks fast because a breath carries a huge concentration, tens of thousands of ppm, so even a small part of it inside the chamber lifts the number a lot; to come down, nearly all of it has to leave, and the approach to the room value gets slower and slower, like any exponential recovery. The fall after the window is opened, which takes whole minutes, is by contrast the airing of the room itself.

Second — and this is a genuinely beautiful connection — **exactly the property that lets the sensor detect CO_2, the absorption of infrared, is what makes CO_2 a greenhouse gas in the atmosphere**. It is worth saying to older or quicker children: *the reason we can measure it is the reason it warms the planet.*

4.2 Why a classroom fills up, and what the thresholds mean

A resting child exhales roughly 0.2 litres of CO_2 per minute; an adult somewhat more. Thirty children in a closed classroom of, say, 180 cubic metres therefore add several litres of CO_2 every minute to a fixed volume of air, and the concentration climbs steadily — by about 1000 ppm in half an hour, so typically from 500 ppm at the start of a morning to 1500–2500 ppm by the end of a closed-window lesson (less if windows and doors let air in). **The rise is close to linear while the room stays shut**, and it is one of the most reliable measurements in the whole platform.

The World Health Organization has no guideline value for indoor CO_2. The European reference is the standard **EN 16798-1**, which sorts the ventilation of buildings into categories by how far indoor CO_2 exceeds the outdoor level: about +550 ppm for category I (very good ventilation), +800 ppm for category II (the normal level for new buildings) and +1350 ppm for

category III. With outdoor air at 420 ppm that means roughly 970, 1220 and 1770 ppm. The app's bands are simpler and rounded, but they say the same thing:

CO_2 level	The app's verdict	What it means
below 800 ppm	**Excellent**	Well ventilated; fresh outdoor air is reaching the room.
800–1200 ppm	**Good**	Acceptable (around categories I–II of EN 16798-1), but the room would benefit from airing.
1200–2000 ppm	**Moderate (air the room soon)**	Insufficient ventilation: the air feels stuffy and many people feel drowsy.
2000–5000 ppm	**Poor (open the windows now)**	Clearly insufficient ventilation; complaints of tiredness and headaches are more frequent.
above 5000 ppm	**Dangerous (open the windows now)**	The 8-hour occupational exposure limit; not reachable in a normal classroom.

Be accurate with children about *why* this matters. **CO_2 at classroom levels is not itself poisonous**, and at 1500 ppm breathing is not "harder". It is a **ventilation marker**: a high reading tells you that the air in the room is largely air that has already been through somebody's lungs, and therefore that fresh air is not arriving. Along with the CO_2 collects everything else people add to the air: moisture, smells and, in the cold season, viruses. That is what makes it a good guide to how often to air the room. Outdoor air, at roughly 420 ppm, is the floor you can never go below indoors.

4.3 The link to plants and to the planet

The app's introduction mentions that plants do the opposite of us: in the light they take CO_2 in. That is the natural bridge to the senior photosynthesis activity. Plants take carbon dioxide, water and light and make sugar and oxygen. At the global scale, the atmospheric concentration has risen from about 280 ppm before the industrial era to over 420 ppm now — which is why the figure 420 in the app is worth pointing at: **it is not an arbitrary number, it is the current composition of the Earth's atmosphere**, and the children can measure something very close to it simply by walking outside with the box.

5. Lesson flow

The in-app wizard has seven steps: an introduction, the prediction, three measurements — the room's air, breathing at the sensor, the window opened —, the **How sure are we? Measure twice** step and a conclusion. Before the measurements the app asks for a **written prediction**, keeps it in the group's journal and quotes it back in the conclusion. Below is what the teacher does around the steps. The times are for the 15-minute short route.

1. **Elicit written predictions first (3 min).** Before a single box is switched on, ask: *if we breathe at the sensor, what will the number do — rise, fall, stay the same? And once we stop, how quickly will it come back?* Almost every child predicts the rise correctly and almost none predicts how *slow* the recovery is. That is where the surprise lives. The class prediction also goes into the app's prediction box.
2. **The introduction on screen (2 min).** Read together what CO_2 is and where it comes from. The activity opens from the **Activities** page with **Start**, and the step ends with **We're ready!**

Technical note — right before the first reading

The sensor. SCD41, a photoacoustic NDIR carbon dioxide sensor that also reports temperature and humidity. The screen shows the **CO_2** card with the unit `ppm`; next to it, the **Air quality:** panel with a coloured bar and the verdict in words (the table in the science background), and underneath the chart over time, with the verdict bands in the background — *Excellent*, *Good*, *Moderate*, *Poor*. In the exported files the column is called `CO2`.

How long you wait — and why the graph climbs in steps. Thirty seconds after power-up, during which the readings are ignored entirely. Then the sensor produces a new value **about every five seconds**, not continuously: the line advances in steps, and this is not a fault. **For the same reason the CO_2 column appears later than the others in the exported file** — the SCD41 is the slowest sensor in the box.

The recovery, not the rise. After a breath the value jumps within seconds, but comes back down over **one to three minutes** — mostly the response time of the sensor itself. Plan that time; it is the most valuable part of the activity.

The firmware. CO_2 is read by firmware 2.2.0 or later. A box still on the old USB-only firmware (`serial_only`) may give no CO_2, and the app says so: if the card stays empty, reflash it with the current WiFi firmware (2.3.0), which works over USB too (the flashing guide is in the `firmware` folder).

The buttons you will use. The green button at the bottom right changes its label at every step, because it repeats what you have just finished: **We're ready!**, **My prediction is written**, **I wrote down the value**, **I observed it!**, **The air is fresh!**, **Understood**, and at the last step the blue **Finish** button. The **Back** button goes back one step without mixing up the steps' notes. At each of the three measurements, the **Capture** button next to the verdict keeps the reading of that moment; the app builds its conclusion from them. If you forget it, a line under the step reminds you, but lets you carry on. The activity is a Junior one, and at this level the app hides the chart's bar (**Smooth**, **Pause**, **Clear**) to keep the children's screen simple. If you want to freeze the shape of the peak on the projector, open the same box on the **Live Data** page as well, where **Pause** is on show. The chart starts at 0:00 at every step, with time in minutes and seconds.

Exporting the data. At the measurement steps, **PNG** (all the step's charts) and **Excel** (the step's raw, unfiltered readings, with an `elapsed_s` column of seconds elapsed) appear. Predictions and notes are saved automatically in the group's journal, in the `logs` folder, and as a **Journal** sheet in the activity's Excel file. **There is no CSV button inside the activity**; files covering the whole lesson are taken from the **Data & Export** page (as `.xlsx` by default). Save following the pattern `class_date_activity` — for example `3B_2026-09-14_co2_respiratie` — and record on the sheet **how many children were in the room and whether the windows were open**, because without them the number cannot be interpreted a week later.

3. **The room's air (3 min).** The box sits on a table in the middle of the room, **away from the children's faces**, and is left to settle. Record the value and the word on the screen. Ask which band on the board the room is in today.

4. **Breathing at the sensor (5 min).** One named child breathes briefly towards the sensor — or, more hygienically, a bottle breathed into earlier is uncapped beside it. Watch the peak. Then **stop and watch the recovery**, with a child reading the number aloud every fifteen seconds. Ask the good question: *why does it come down so slowly, when it went up so fast?* An answer at their level: the sensor is like a little room with a small door; the breath gets in quickly because there is so much of it, but for all of it to get out takes time, and the sensor only looks once every five seconds.

5. **The window opened, and the conclusion (2 min).** Open the window and watch the fall — this is the airing of the room itself. The next step, **How sure are we?**, asks for a second

breath, the same way, once the number has settled; on the short route just read it aloud, and do the second breath in lesson 1. Fix the conclusion in the children's own words: the air we breathe changes when there are many of us in a closed room, and it comes right again when we open the window. Finish with the app's conclusion step, which shows them their written prediction again and whose blue button reads **Finish**, and ask every child to complete the sentence on their sheet.

If you have twenty minutes more

Three extensions, in rising order of interest. **The fan**: move air over the sensor with a piece of card and watch the recovery speed up a little — ventilation is simply moving air. **The jar**: the box closed alone inside a large jar shows a *flat* line — nothing inside breathes, so nothing changes the CO_2. It is the experiment's control. **If the line climbs anyway, the jar is not airtight** and the room's air, filling up meanwhile, is getting in. With *a plant* inside and in the light the line falls slowly, and the discussion changes completely. **The air map**: the corridor, the hall, outdoors by the door and the classroom, measured in turn and ranked — outdoors will always be the cleanest, around 420 ppm, and the children will demand to know why.

The investigation, in one table

The question. Does the air in our room change when we breathe in it?
The hypothesis. Written on the sheet **before** any measuring, with a "rises / falls / stays the same" for the breath and an opinion about how quickly the number comes back. It is not changed afterwards.
What I change. One thing only: the **air** around the sensor — first the room's, then exhaled air, then the air outside.
What I measure. Carbon dioxide, in ppm, as shown on the card, **and the time** until the value comes back close to where it started.
What I keep the same. The same box, in the same place, on a surface and not in a hand; the same distance to breathe from; **nobody else leaning over it** — and this is the rule broken most often, because every child wants to see.
How many repeats. Two breaths, with a full recovery between them. If the second gives a peak close to the first, the measurement is good; if the second starts much higher, the recovery was not waited for.
What makes a reading valid — and what makes it an artefact. Valid is a reading taken with the box on the table, nobody above it, at least thirty seconds after power-up. An artefact is anything else: **a ring of children leaning over the box raises the room value by hundreds of ppm** without anyone breathing at it deliberately, and then "the room's value" is not the room's but the ring's. The class rule: we measure from a step back.

6. Analysis after the activity

6.1 What the data should look like

Situation	Typical value	What it means
Outdoors	400–450 ppm	The atmosphere's value; the floor below which you cannot go indoors.
Aired room, start of the day	450–800 ppm	Excellent.
Room with 30 children, after a closed hour	1500–2500 ppm	Close to a linear rise; airing recommended.
Corridor	600–1000 ppm	Usually better than the classroom: more volume, fewer people.
Just after a breath at the sensor	2000–5000 ppm	A sharp peak; back down in 1–3 minutes.
Exhaled air, directly	about 40 000 ppm	Which is why nobody breathes towards the box when the room is being measured.
Five minutes after opening the window	a fall of 300–800 ppm	Fast at first, then slower and slower.

Absolute values depend on how many children there are, how large the room is and how well the windows seal. Do not look for a "correct" number. What must come out reliably is **the shape**: a slow climb while the room is shut, a sharp peak on a breath, a slow recovery, a fall on airing.

6.2 Questions to ask afterwards

1. The number went up in a second when we breathed on it, but took a minute and a half to come down. Why do you think it needed more time to come back?
2. The air outdoors read 420 ppm. Why can we never get *below* that in the room, however much we air it?
3. If there were twenty of us tomorrow instead of thirty, what would happen to the graph over the lesson?
4. The sensor showed temperature and humidity rising at the same instant as the CO_2. Why do all three go up together when you breathe?
5. When should the window be opened in our classroom, and who should look after it? Use your numbers to support your rule.

6.3 Assessment ideas

- **Low stakes:** the child circles the right word on the sheet — "Good", "Moderate", "Poor" — for each moment, and draws the shape of the line: the sudden rise and the slow descent. No numbers required.
- **Standard:** the child records the three values — the room, the peak, after airing — finds the difference by subtraction, and writes a full sentence saying where the room's CO_2 comes from.
- **Extension:** the child times the recovery, compares it with another group's and explains why they differ; or predicts the corridor value before measuring it and explains the mismatch.

Common pitfalls and misconceptions

The ring of children leaning over the box. The number one cause of strange data. Ten children leaning in to see the screen raise the value by hundreds of ppm in under a minute,

and then "the room's air" is really the ring's air. Measure from a step back, with the screen projected or a named reader.

Rushing the recovery. A child who breathes a second time before the first recovery has finished gets a peak that starts high and looks smaller. Wait for the full recovery.

"CO_2 is poison." It is not, at classroom levels, and **children must not be frightened**. It is an indicator: it shows that the air has already been through somebody's lungs and that fresh air is not arriving. Put it exactly that way.

Reading in the first thirty seconds. The values at power-up are not trustworthy. Ignore them entirely.

Expecting a smooth line. The sensor gives one value every five seconds, so the line climbs in steps. It is not broken.

Mistaking the sensor for a "bad air" detector. It does not see dust, smoke or smells — only CO_2. A room can smell unpleasant and have low CO_2, or the other way round.

A box that reads well below 400 ppm or above 500 ppm outdoors. Outdoor air, away from cars and people, is at 420–450 ppm. A large offset means the sensor has lost its calibration (its automatic self-calibration assumes it sees fresh air now and then); the technical manual shows how to recalibrate it outdoors. For this activity the differences matter, not the exact value.

Opening the window in a gale. The fall is so fast that the shape cannot be seen, and the papers fly. On a windy day open one window, and only a little.

Safety and classroom management

Hygiene when breathing at the sensor. Decide the rule *before* the lesson and state it clearly. The safest version: **nobody breathes directly at the box**; instead each child breathes into their own plastic bottle, which is then uncapped beside the sensor. If anyone does breathe directly, it is one named child, from a hand's distance, and the box is wiped afterwards. **Children with a cold do not breathe at it at all** — they take the screen-reader or timekeeper role instead.

Windows are opened by the teacher, not by children. Nobody climbs on a sill, a radiator or a chair to reach the handle. If the window opens onto a yard with people in it, check before opening.

Balloons. A burst balloon startles, and with young children pieces of rubber are dangerous if they reach the mouth. Blow the balloons up yourself, collect any burst one immediately, and do not let young children put them in their mouths.

Going outdoors for the reference reading follows the school's usual rules: count the children out and back, and set the boundaries before you leave.

Cables. A USB cable stretched to the window is both a trip hazard and a way to pull the laptop off the table. Use WiFi where you can, or tape the cable down.

Adaptations for pupils with special educational needs

No large numbers at all. This is the main adaptation. The child records no four-digit values: they circle the word from the screen — "Good", "Moderate", "Poor" — for each moment. The learning objective is that the air changes when we breathe in it and that a machine can tell us so, and it is met in full with no large figures.

Drawing the shape instead of filling a table. The sheet has a box in which the child draws the line: up fast, down slowly. **For many children the shape is a better representation of the phenomenon than three numbers**, and it can be discussed just as well.

With a single box, the whole class. Semicircle, screen projected, and you handle the box. At each moment name a **screen reader**, who says the number aloud, and a **timekeeper**,

who announces how much of the recovery has passed.
Three real roles for children with motor difficulties. Not being able to carry the box does not mean not taking part. The roles rotate:

- **the timekeeper** — starts the clock at the breath, announces every fifteen seconds and says when the value has come back; this is the role with the greatest scientific value in the group, because they measure the very quantity that surprises the class;
- **the screen reader** — says the number aloud, rounded;
- **the drawer** — traces the shape of the line on the sheet, in real time.

Children who cannot or will not breathe at the sensor. Asthma, a cold, a sensory sensitivity, or simply refusal — all are sufficient reasons. **The bottle with a kept breath solves the problem completely**: the teacher can provide the breath, and the child uncaps the bottle, which is the very gesture that starts the experiment.
Language and patience. Use the same three words all lesson (carbon dioxide, fresh air, airing) and announce in advance that the number **will come back slowly** — otherwise an anxious child seeing a large value thinks they have broken something or that the air is dangerous.

If things do not go to plan

You have one box and thirty children. Semicircle, screen projected, and you handle the box. This is in fact the recommended version here, because it **keeps the children away from the sensor**, which has to be done anyway.
You have fifteen minutes. This is the short route. Drop the air map and the jar, but **do not drop the recovery** — it is the lesson.
The window will not open, or there is a gale outside. Use the cardboard fan: moving air over the sensor speeds up the recovery and shows exactly the same thing on a small scale. Or take the box into the corridor for ten seconds and bring it back.
The room value is already very high at the start of the lesson. Not a problem but a discovery: the room was not aired. Begin the lesson by airing it and watch the fall — the order of the experiments changes, the content does not.
The number does not move at all. Check first whether the thirty seconds since power-up have passed. The SCD41 gives a value every five seconds, so between readings the line genuinely does stand still. **The CO_2 card stays empty altogether?** The box probably runs the old USB-only firmware (`serial_only`) older than 1.3.4, which does not read CO_2 (the app says so): reflash it with the current WiFi firmware.
The box will not connect. Do not stop the lesson. Use the "What the data should look like" table as a printed dataset: the children write their predictions first, then receive the values row by row and check whether they guessed the order. The discussion about why the recovery is slow is unchanged, and the decision part — when do we open the window — can be done entirely without the equipment. Measure again another day, after checking the cable, the port selected in the app and, if you work over WiFi, the network.

7. Further resources

Stable, well-established starting points. Search within each site for the topic named — these are base addresses, not deep links, so they do not rot.

- **World Health Organization** — `https://www.who.int` — search for indoor air quality; its guidelines cover pollutants (particles, combustion gases, damp), **not CO_2**, for which the WHO gives no guideline value.
- **REHVA** — `https://www.rehva.eu` — the European federation of ventilation engineers; search for EN 16798-1 and the guidance on ventilating schools, where the categories above

come from.

- **PhET Interactive Simulations** — `https://phet.colorado.edu` — search for the gases and greenhouse effect simulations; the second links the classroom measurement directly to the atmosphere.
- **STEM Learning** — `https://www.stem.org.uk` — search for respiration and air for ages 7–11; a large library of lesson plans this measurement slots straight into.
- **Institute of Physics** — `https://www.iop.org` — search their primary material on gases and the properties of air, written for teachers who are not specialists.
- **NASA Climate** — `https://climate.nasa.gov` — search for atmospheric carbon dioxide concentration, if you want to show the class where the 420 ppm figure comes from and how it has changed.
- **Royal Society of Biology** — `https://www.rsb.org.uk` — for the link to respiration and plants, if you go on to the jar extension.

8. Assessment

The exit ticket, identical to the one on the pupil sheet: *one thing that surprised you today.* Collected in the last two minutes of the lesson, it says more about the class's understanding than any mark.

For a more formal judgement, use the **common three-axis rubric** — *measurement practice*, *data interpretation* and *scientific reasoning.* At Junior level the three axes look like this:

- **Measurement practice** — the child stays back from the sensor when the room's air is being measured, waits for the full recovery between breaths, and records what the screen shows rather than what they expected.
- **Data interpretation** — the child says which band the room is in, notices that the peak rises fast and falls slowly, and compares the value before airing with the one after.
- **Scientific reasoning** — the child explains, in their own words, where the room's CO_2 comes from and why opening the window lowers it, and compares their conclusion with the prediction written at the start.

The rubric itself, with descriptors for all three levels, is defined once in the *curriculum plan* and is not repeated in the guides.

9. Pupil sheet

The sheet the children fill in is a separate document, the file `docs/en/student_sheets/air_we_breathe.pdf`. It has two pages — the prediction, the threshold scale and the drawing box on the first; the measuring, the recovery time and the conclusion on the second — so it is printed double-sided, one sheet per child. It is built on the three layers — what I see on the screen, what I do on my own, the conclusion I write — and it asks for the prediction **before** any measuring.

Three things to know before printing it. **The sheet asks for the recovery time, not just the peak**, with a box of its own: the recovery is the quantity that surprises the class, and if it has no place on the paper nobody measures it. **The box for drawing the shape of the line comes before the table**, not after it, because drawing is the very task that fills the one to three minutes of waiting. **The column with the words from the screen** — "Good", "Moderate", "Poor", the bands in the chart's background — is there precisely for the child who does not work with four-digit numbers: they circle it and leave the number column empty. The table is not repeated here precisely so that the two cannot drift apart: there is one printable version, and it is the sheet.